Robust Multichannel Perron Envelopes
Target 56 of the H1 Downstream Programme

Yang–Mills representational dynamics research programme

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Abstract

We replace a single calibrated defect kernel by a robust entrywise envelope. The Perron vector of the envelope controls every dominated physical candidate. For the four-channel H1 star we calculate the exact additive, multiplicative, and symmetric persistence/tunneling slack before the degree-18 cluster threshold. The optimization problem is closed; constructing the complete nonlinear channel envelopes remains open.

1 Envelope theorem

Let Q, B be finite nonnegative matrices with $Q \le B$ entrywise. Suppose B is irreducible and $BV_B = \rho(B)V_B$ with $V_B > 0$.

Theorem 1 (Robust Perron domination)**.**

$$QV_B \le \rho(B)V_B.$$

Consequently every kernel in the order interval $[0, B]$ *has weighted ceiling at most* $\rho(B)$ *in the common weight* V_B*.*

Proof. Positivity gives $QV_B \le BV_B$ componentwise. The conclusion follows from the Perron equation. □

This theorem is deliberately one-sided. A numerically plausible row not proved to lie below B receives no certificate.

2 Four-channel uncertainty budget

Let $k = 4$ and use the root/charged matrix

$$B_{0a} = \frac{q^2}{k}, \qquad B_{a0} = 1 - s, \qquad B_{aa} = p, \qquad B_{ab} = \frac{c}{k-1} \quad (a \ne b),$$

where $s = p + c$. On the symmetric charged subspace the Perron root is

$$\rho(s) = \frac{s + \sqrt{s^2 + 4q^2(1-s)}}{2}. \tag{1}$$

For $s = 0$, $\rho = q = 0.002430277501841998$. With $q_{\mathrm{crit}} = 0.0027927080803578997$, the exact budgets are

$$\begin{aligned} q_{\mathrm{crit}} - q &= 0.0003624305785159018, \\ \frac{q_{\mathrm{crit}}}{q} - 1 &= 0.1491313556748981, \\ s_{\max} &= \frac{q_{\mathrm{crit}}^2 - q^2}{q_{\mathrm{crit}} - q^2} = 0.0006792623948561134. \end{aligned}$$

The last number is the total symmetric charged-to-charged persistence or tunneling mass compatible with strict subcriticality. At half this budget the certified radius is approximately $0.00260560708 < q_{\mathrm{crit}}$.

Corollary 1 (Additive weighted errors)**.** *If $BV \le qV$ and $EV \le \varepsilon V$, then $(B + E)V \le (q + \varepsilon)V$. Hence every normalization, chart-complement, and cross-channel error consumes the same additive weighted slack.*

3 Physical channel ledger

The required envelope must include continuous incoherence, center frustration, cut/orbit charts, compositionally complete integer topology, persistence, and tunneling. Entrywise domination is sufficient, but a directly weighted bound may be sharper. Either route must use normalized conditional rows uniformly over positive-mass histories.

Warning 1. *The finite envelope is not a physical Yang–Mills kernel. Omitting a channel, using an unnormalized numerator, or accepting equality at q_{crit} invalidates the cluster conclusion.*

4 Audit

The robust theorem and budgets are C2. Eleven hostile tests verify spectral formulas, threshold strictness, domination, cross-channel equivalence, invalid weights, and superprobability rejection.